

Name: _____

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- [illegible]

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- This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

3. (5 points) We want to compute the number of occurrences of each word in a file. Outline a solution using the MapReduce framework (pseudocode only) to compute word frequencies.

This image shows a full page of primary-ruled notebook paper. It features ten sets of horizontal lines across the page. Each set consists of three lines: a solid top line, a dashed middle line, and a dotted bottom line. The lines are evenly spaced and extend from the left margin to the right edge of the page. There is no handwriting or other markings on the paper.

4. (5 points) Give pseudocode for computing a join $r \bowtie_{r.A=s.A} s$ using a single **MapReduce** step, assuming that the *map()* function is invoked on each tuple of r and s . Assume that the *map()* function can find the name of the relation using *context.relnam()*.

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